

Miscellaneous Slides Deleted
from Main Presentation

Log Likelihood Values for different GARCH models

	ARCH(4)	GARCH(1,1)	GARCH(1,4)	GARCH(3,4)
log Lik.	2712.45	2712.29	2724.52	2728.04

UPDATE THE ABOVE TABLE. That is copy from ETHEREUM table.

Explanation: We can see the maximized log likelihood of ARCH(4) and GARCH(1,1) are close to each other, and GARCH(1,1) has less fitted parameters than ARCH(4) and AIC favors GARCH(1,1), which shows that GARCH(1,1) performs well as a popular model. GARCH(1,4) and GARCH(3,4) have larger log likelihood, which is not surprising since they have more parameters.

Update this language. And based on the best GARCH model use that GARCH model to show all the graphs as are shown in Ethereum project.

Key Findings from EGARCH Model - 3

Weighted Ljung-Box Test and ARCH LM Tests on Squared Residuals: Both of these tests check for remaining ARCH effects. In both cases, the p-values are large (e.g., Ljung-Box p-value at lag 1 is 0.6145; ARCH LM p-value at lag 3 is 0.2234). We do not reject the null hypothesis. This is a success. The EGARCH model has successfully accounted for the conditional heteroskedasticity (volatility clustering) in the data.

Sign Bias Test: The sign bias test the presence of leverage effects in the standardized residuals (to capture possible misspecification of the GARCH model), by regressing the squared standardized residuals on lagged negative and positive shocks. The Sign Bias test regresses a dummy for positive and negative sign bias on the squared residuals

The null hypothesis is not rejected as the calculated t-value is lower than the critical value from t-table. It implies that both positive and negative news do not have a significant impact on the price of Bitcoin.

This test checks if any uncaptured asymmetric effects remain. The p-values for all terms are large (e.g., Joint Effect p-value = 0.9955). We do not reject the null hypothesis. This is a key success, confirming that the EGARCH model has effectively captured the asymmetric nature of Bitcoin's volatility response.

Null Hypothesis Stability Test: This tests for parameter stability over time. The Joint Statistic (1.7572) is between the 10%

Key Findings from GARCH Model - 1

Mean Model: This is an ARMA(1,1) model for the returns. It models the current return based on the previous day's return ($ar1$) and the previous day's forecast error ($ma1$).

GARCH Model: Standard GARCH(1,1) model for the variance. It models today's variance based on yesterday's squared error (α) and yesterday's variance ($\beta1$).

Mean Equation: The $ar1$ coefficient is not statistically significant ($p\text{-value} = 0.122103 > 0.05$). The $ma1$ term is significant ($p\text{-value} = 0.017870 < 0.05$).

Variance Equation: All parameters (ω , $\alpha1$, $\beta1$) are statistically significant ($p\text{-values} \neq 0.000000$). This confirms the presence of GARCH effects.

Volatility Persistence: The sum of $\alpha1 + \beta1$ ($0.077659 + 0.921341 = 0.999$) is almost 1. This indicates that shocks to volatility are persistent and it takes long time for volatility to return to its long-run average after a shock.

GARCH Model Introduction

The GARCH Model or the Generalized Autoregressive Conditional Heteroskedasticity is a widely used model for financial time series modeling and the model for GARCH takes the following form: $\text{GARCH}(p,q)$. The GARCH model allowed us to capture a couple of fundamental properties of the GARCH model, this includes both volatility clustering and persistence. The parameters of the model were significant for volatility clustering which supported the fact that periods of high volatility are followed by more high volatility. In volatility persistence, the sum of the alpha and beta coefficients was almost 1 which shows us that shocks in the Bitcoin's volatility are persistent. The GARCH model was crucial as by helping fit a more standard model, more diagnostic tests helped better understand what the model is lacking. The GARCH model used the Sign Bias Test results to state that the analysis performed on its residuals was able to reveal highly significant uncaptured asymmetries. This allowed us to understand that the sGARCH model would not let us meet our data analysis goals which is why we need to move onto a more complex model with EGARCH or GJR-GARCH analysis.

ARIMA test - THIS SLIDE NOT UPDATED. This includes info when data was not changed to log scale

Brief Description of the ARIMA Model: I model using the ARIMA test to calculate AR: Autoregressive. This component calculates the relationship between a current observation and its past values. For an ARIMA(5,1,0) model, it calculates five coefficients (ar1 through ar5) that quantify how much the five previous time steps influence the current value. This component calculates the level of differencing required to make the time series stationary (i.e., to have a constant mean and variance over time). Differencing involves subtracting a past value from a current value. A d=1 means the model calculates the difference between residual errors from past forecasts. An MA(0) model indicates that this component is not used.

```
> modelrBTC_Close=arima(rBTC_Close, order = c(5,1,0))  
> modelrBTC_Close
```

Call:

```
arima(x = rBTC_Close, order = c(5, 1, 0))
```

Term	Coefficient	Standard Error (s.e.)	t-statistic
ar1	-0.8969	0.0064	-140.1
ar2	-0.6686	0.0083	-80.5
ar3	-0.4902	0.0088	-55.7
ar4	-0.3314	0.0083	-39.9
ar5	-0.1220	0.0064	-19.1

auto.arima (rBTC_Close). Using that function, I get the following results:

The ARIMA Model of has the format of ARIMA(p, d, q) which stands for:

What does the p=5 mean: This is the AR(5) part or the autoregressive. The model assumes the current value is a linear combination of the

What does the d=1 mean: This is the I(1) part and is critically important. Stationary, the data has been differenced once (current_value - pre model is essentially analyzing the change in returns, or the "acceleration").

What does q=0 mean: This is the MA(0) part and is critically important. The data has been differenced once (current_value - previous_value). This is essentially analyzing the change in returns, or the "acceleration" of the Bitcoin price. This implies that the returns series itself was found to be non-stationary (a highly significant finding that should be investigated with a unit root test like the Augmented Dickey-Fuller test).

Coefficients:

	ar1	ar2	ar3	ar4	ar5
	-0.8969	-0.6686	-0.4902	-0.3314	-0.1220
s.e.	0.0064	0.0083	0.0088	0.0083	0.0064

sigma^2 estimated as 197.3: log likelihood = -97975.31, aic = 195962.6